LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development
CourseCode: 23CM4121
ExperimentNo: 1&2
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1. TITLE

Implementation of JavaScript Arrays and Functions

2. AIM

To understand and implement **arrays, for-of loops, functions, parameters, return values, and console output** using JavaScript

3. OBJECTIVES

The objectives of this experiment are:

- To understand the concept of arrays in JavaScript.
- To create and access elements of an array.
- To use the **for-of loop** to traverse array elements.
- To understand the concept of functions in JavaScript.
- To pass values as function parameters.
- To return a value from a function.
- To display the output using console.log().
- To execute and verify the JavaScript programs.

4. THEORY

JavaScript is a programming language commonly used to add logic and interactivity to web pages.

An **array** is a collection of multiple values stored in a single variable. In the given program, the array contains three fruit names: **Apple, Banana and Mango**.

The **for-of loop** is used to access each element of an array one by one.

A **function** is a reusable block of code that performs a specific task. In the second program, the `add()` function accepts two parameters, adds them, and returns the result.

The `console.log()` method is used to display output in the browser's **Developer Console**.

Basic flow:

JavaScript Code → Execution → Processing → Console Output

5. ALGORITHM / PROCEDURE

Program 1 – Display Array Elements

1. Start the program.
2. Create an array named fruits.
3. Store "Apple", "Banana" and "Mango" in the array.
4. Display "Fruits:" using console.log().
5. Use a for-of loop to access each fruit.
6. Display each fruit using console.log().
7. Stop the program.

Program 2 – Addition Using Function

1. Start the program.
2. Define a function named add().
3. Pass two parameters a and b to the function.
4. Add the two parameters.
5. Return the calculated value.
6. Call the function using add(10, 20).
7. Store the returned value in the variable result.
8. Display the result using console.log().
9. Stop the program.

6. PROGRAM – Array.js

Program 1: Display Fruits Using Array and For-of Loop

```javascript
let fruits = ["Apple", "Banana", "Mango"];

console.log("Fruits:");

for (let fruit of fruits) {
    console.log(fruit);
}
```

6. PROGRAM – Function.js

Program 2: Addition Using Function

```javascript
function add(a, b) {⋯
}

let result = add(10, 20);

console.log("Sum =", result);
```

7. CODE EXPLANATION

Code	Explanation
let fruits = ["Apple", "Banana", "Mango"];	Creates an array named fruits and stores three fruit names.
console.log("Fruits:");	Displays the heading "Fruits:" in the console.
for (let fruit of fruits)	Traverses each element of the fruits array one by one.
console.log(fruit);	Displays each fruit name in the console.
function add(a, b)	Defines a function named add with two parameters.
return a + b;	Adds a and b and returns the result.
let result = add(10, 20);	Calls the add() function with 10 and 20 and stores the result.
console.log("Sum =", result);	Displays the calculated sum in the console.

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8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Execute the fruits program	Apple, Banana and Mango should be displayed	Apple, Banana and Mango displayed	Pass
TC-02	Execute the addition program with 10, 20	Sum should be 30	Sum = 30	Pass
TC-03	Execute the fruits program	Fruits heading should be displayed	Fruits: displayed	Pass

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-04	Call add(10, 20)	Function should return 30	30 returned	Pass
TC-05	Open browser Developer Console	JavaScript output should be visible	Output visible in Console	Pass

Expected Console Output

```
Fruits:
Apple
Banana
Mango

Sum = 30
```

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9. OUTPUT

Output 1 – Array Program

```
Fruits:
Apple
Banana
Mango
```

The program successfully displays all elements of the `fruits` array.

Output 2 – Function Program

```
Sum = 30
```

The `add()` function successfully adds `10` and `20` and returns `30`.

Output Screenshots:
Insert the actual browser **Developer Console screenshot** showing the above output here.

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10. RESULT

Thus, the JavaScript programs using **arrays, for-of loops and functions** were successfully implemented and executed. The array elements were displayed correctly, and the function successfully calculated the sum of two numbers as **30**.

```
PS C:\Users\AISHWARYA\OneDrive\Desktop\clg information portal> node array.js
Fruits:
Apple
Banana
Mango
PS C:\Users\AISHWARYA\OneDrive\Desktop\clg information portal> node function.js
Sum = 30
PS C:\Users\AISHWARYA\OneDrive\Desktop\clg information portal>
```